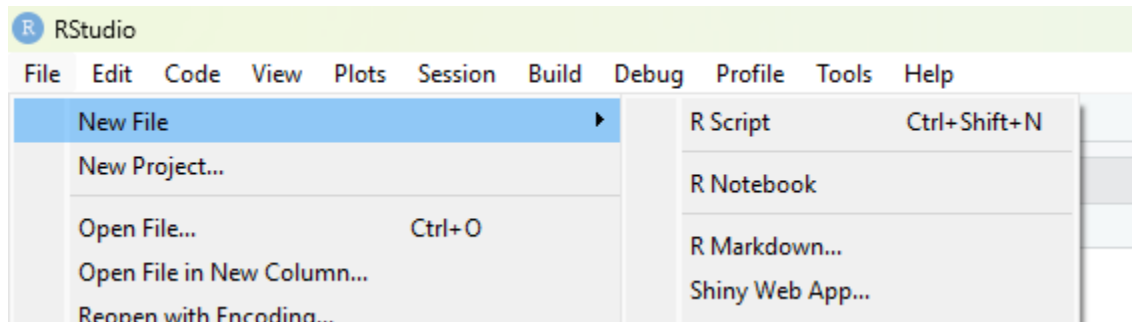


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Practical 5

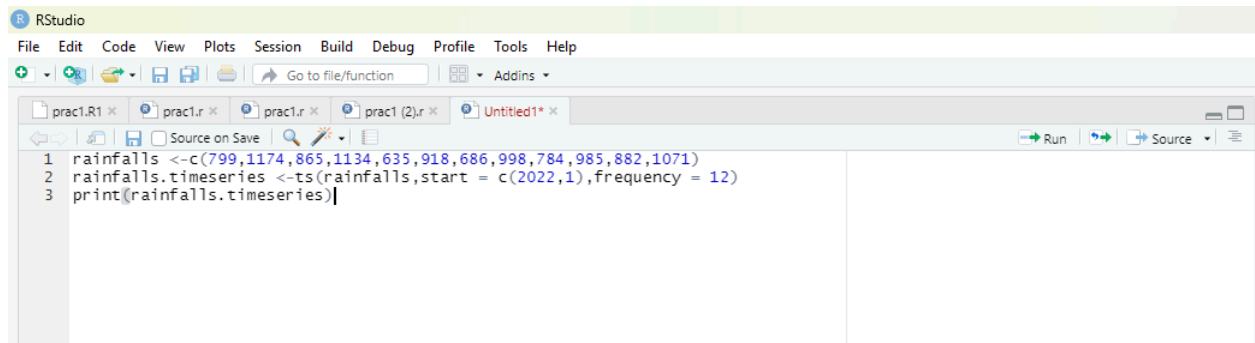
Aim : Perform the data classification using classification algorithm.

1. Open Rstudio and click new script



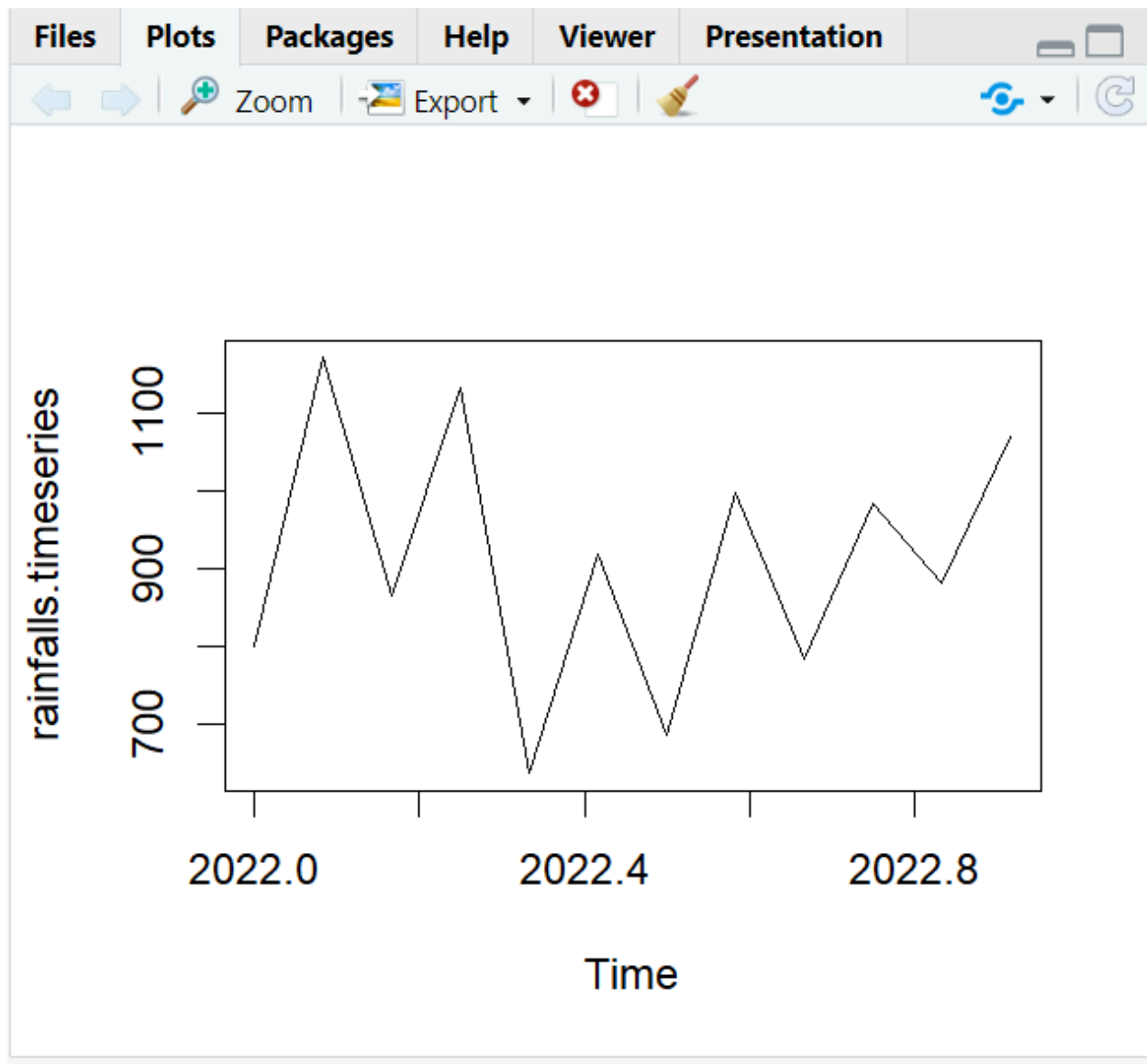
2. Type code and run

```
rainfalls <-c(799,1174,865,1134,635,918,686,998,784,985,882,1071)
rainfalls.timeseries <-ts(rainfalls,start = c(2022,1),frequency = 12)
print(rainfalls.timeseries)
```



Second Code

```
png(file="rainfall")|
plot(rainfalls.timeseries)
dev.off()
plot(rainfalls.timeseries)
```



Practical 6

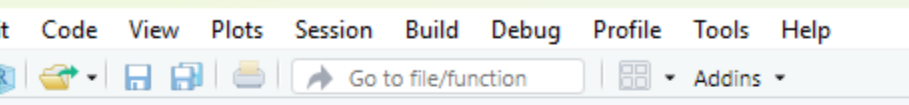
Aim : K-Means clustering using R

1. First code

```
newiris <- iris
```

```
newiris$Species <- NULL
```

```
kc <- kmeans(newiris,3)
```



The screenshot shows the RStudio application window. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu bar is a toolbar with icons for creating a new file, opening a file, saving, and other functions. The main editor area displays three tabs: Untitled1*, Untitled2*, and Untitled3*. The active tab is Untitled3*, which contains the following R code:

```
1 newiris <- iris
2 newiris$Species <- NULL
3 kc <- kmeans(newiris,3)
4 kc
```

[illegible]

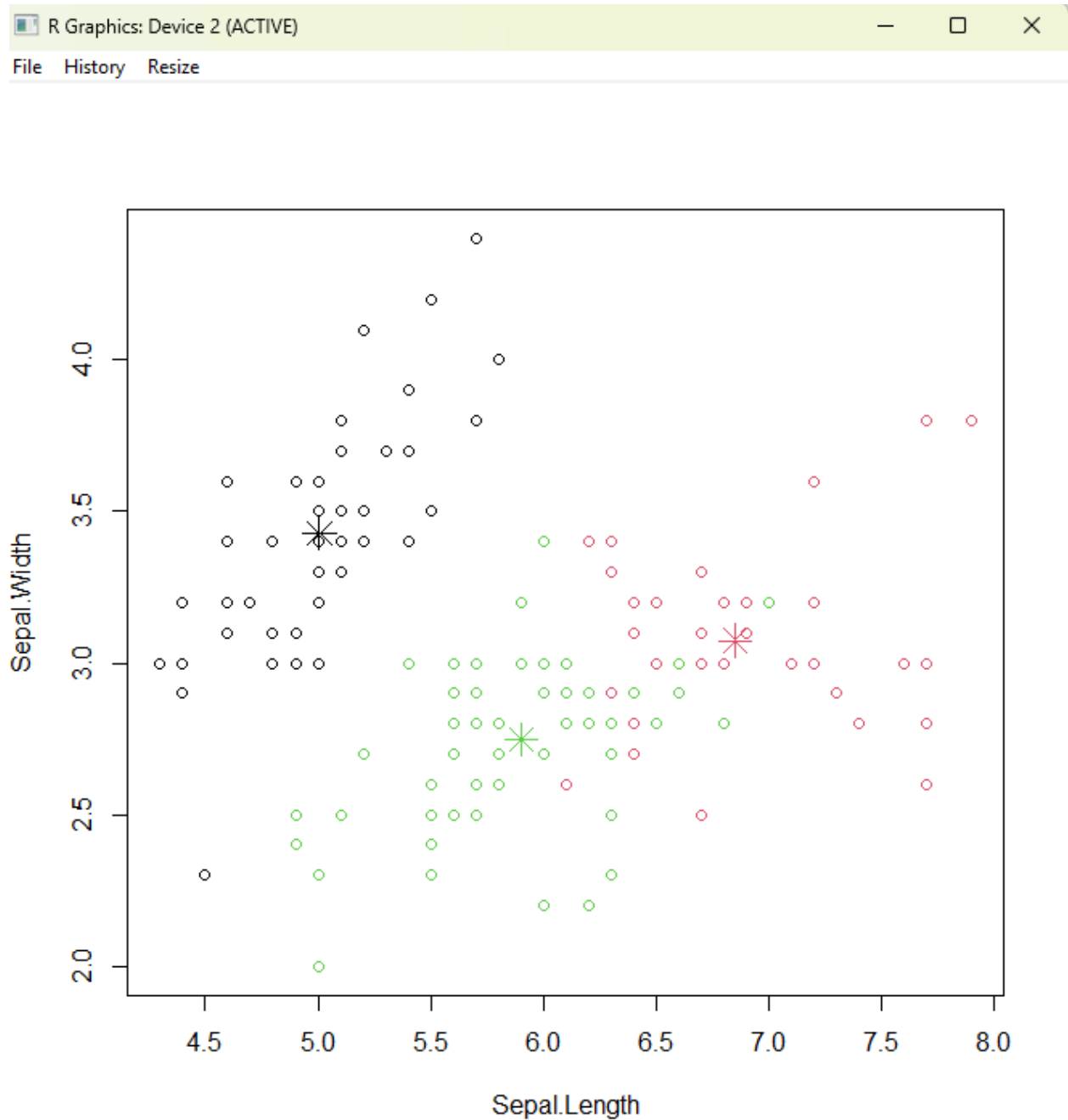
Output

	1	2	3
setosa	17	0	33
versicolor	4	46	0
virginica	0	50	0

3. Third code

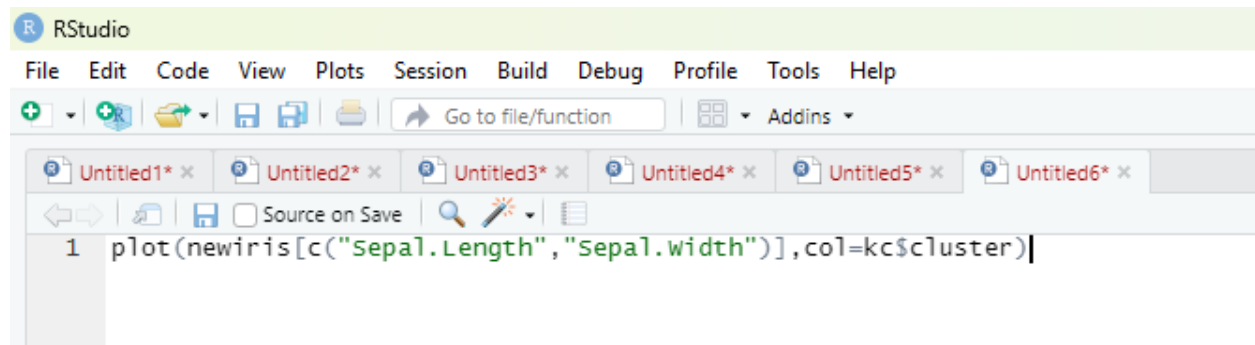
```
plot(newiris[c("Sepal.Length", "Sepal.Width")], col=kc$cluster)
points(kc$centers[,c("Sepal.Length", "Sepal.Width")], col=1:3, pch=8, cex=2)
dev.off()
```

Output:-



4. Fourth code

```
plot(newiris[c("Sepal.Length", "Sepal.Width")], col=kc$cluster)
```



Practical 7

:AIM: Predict using Linear Regression

Step1:

Code1

```
x<-c(151,174,138,186,128,136,179,163,152,131)
```

```
y<-c(63,81,56,91,47,57,76,72,62,48)
```

```
relation<-lm(y~x)
```

```
print(relation)
```

```
> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> print(relation)
```

```
Call:
lm(formula = y ~ x)
```

```
Coefficients:
(Intercept)          x
   -38.4551      0.6746
```

Step2:

Code2

```
x<-c(151,174,138,186,128,136,179,163,152,131)
```

```
y<-c(63,81,56,91,47,57,76,72,62,48)
```

```
relation<-lm(y~x)
```

```
print(summary(relation))
```

```
> x<-c(151,174,138,186,128,136,179,163,152,131)
> y<-c(63,81,56,91,47,57,76,72,62,48)
> relation<-lm(y~x)
> print(summary(relation))
```

```
Call:
lm(formula = y ~ x)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-6.3002 -1.6629  0.0412  1.8944  3.9775
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -38.45509    8.04901  -4.778  0.00139 **
x             0.67461    0.05191  12.997 1.16e-06 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3.253 on 8 degrees of freedom
Multiple R-squared:  0.9548,    Adjusted R-squared:  0.9491
F-statistic: 168.9 on 1 and 8 DF,  p-value: 1.164e-06
```